Chapter 2: Boolean Algebra and Logic Gates

MULTIPLE CHOICE

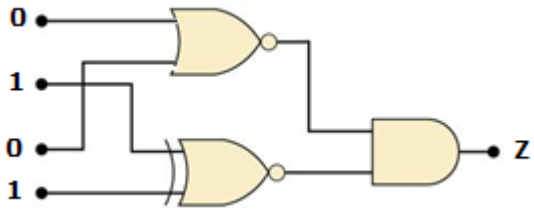
1. An exclusive-OR function is expressed as.
   a) A'B'+AB
   b) A'B+AB'
   c) (A'+B)(A+B')
   d) (A'+B')+(A+B)

2. Which one of the following is not a valid rule of Boolean algebra?
   a) A+1=1 b) A=A'
   c) AA=A d) A+0=A

3. An example of product-of-sums expression is?
   a) A(B+C)+AC'
   b) (A+B)(A'+B+C')
   c) A'+B'+BC
   d) both answers (a) and (b)

4. Given that F= X Y Z' + X' Y. Then F. F'. F' equals
   a) 1
   b) X' Y + X' Y Z + X Y Z' + X Y' Z
   c) 0
   d) X' Y + X Y Z'

5. What is the value of *Z* in the shown figure that has the shown bits at its inputs?
   a) 0
   b) 1
   c) Either 0 or 1
   d) No correct answer



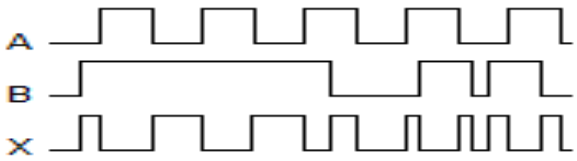
6. Simplifying the logic expression $(AB + \overline{AB})(AB + ABCD)$ gives:
   a) 1 b) A' + B'
   c) AB d) (AB)'

7. According to the De Morgan's theorems, the following equality(s) is (are) correct:
   a) $\overline{A+B+C} = \overline{A}\,\overline{B}\,\overline{C}$
   b) $\overline{AB} = A + \overline{A}\,B$
   c) $\overline{X} + \overline{Y} + \overline{Z} = (\overline{X} + \overline{Z})(\overline{X} + \overline{Y})$
   d) All of the above

Answers : B B B C B C A

8. The following waveform pattern is for a(n) …… (A and B are inputs, X is an output).

A. 2-input AND gate
B. 2-input OR gate
C. Exclusive-OR gate
D. None of the above



9. By using the Boolean algebra techniques,
Simplify the following expression

$$f = xy\bar{z} + \bar{x}yz + xyz + \bar{x}y\bar{z}$$

A. $x\bar{y} + xz$
B. $(x + y + z)$
C. y
D. $\bar{x}$

10. The Boolean expression x + yz is equal to which of the following Boolean expressions?

A. x + y + z
B. x. (y + z)
C. (x + y). z
D. (x + y) (x + z)

11. To implement the expression $A\bar{B}CD + AB\bar{C}D + ABC\bar{D}$, we need one OR gate and ……

A. three AND gates and three inverters
B. three AND gates and four inverters
C. three AND gates
D. one AND gate

12. A gate produces a HIGH output whenever its two inputs are the same.

A. Exclusive-AND
B. Exclusive-NAND
C. Exclusive-OR
D. Exclusive-NOR

13. According to the De Morgan's theorems, the following equality(s) is correct:

A) $\overline{AB} = A + \overline{A}B$
B) $\overline{X} + \overline{Y} + \overline{Z} = (\overline{X} + \overline{Z})(\overline{X} + \overline{Y})$
C) $\overline{A + B + C} = \overline{A}\,\overline{B}\,\overline{C}$
D) $\overline{A + B + C} = \overline{A} + \overline{B} + \overline{C}$

14. Which of the following gates are universal gates?

A) XOR and XNOR
B) NAND and NOR
C) NAND and XNOR
D) NOR and XOR

15. Given that F= X Y Z' + X' Y. Then $F + \bar{F}$ equals

A. 1
B. X' Y + X' Y Z + X Y Z' + X Y' Z
C. 0
D. X' Y + X Y Z'

Answers : C C D A D C B A

16. Express the following function as a sum of product SOP minterms form.

A. $\overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}BC$

B. $\overline{A}\overline{B}\overline{C} + A\overline{B}C + \overline{A}BC$

C. $\overline{A}\overline{B}\overline{C} + \overline{A}BC + AB\overline{C}$

D. None of the above.

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Answers : C

17. Prove xy+yz+zx=(x+y)(y+z)(z+x) ?